

Linux Task 03: Process Management, System Monitoring & Basic Shell Scripting

Part A: Process Monitoring

What is a Process?

- A process is a running instance of a program.

What is a PID?

- PID stands for Process ID, a unique number assigned to each running process.

Which process is consuming the most CPU?

- Xorg.

Which process is consuming the most Memory?

- Xorg with PID 689

```
hacker@hacker:~/task1/mytask
(hacker@hacker)~$ top
top - 08:12:28 up 41 min, 1 user, load average: 0.16, 0.20, 0.17
Tasks: 174 total, 1 running, 173 sleeping, 0 stopped, 0 zombie
%Cpu(s): 5.5 us, 3.7 sy, 0.0 ni, 89.9 id, 0.0 wa, 0.0 hi, 0.9 si, 0.0 st
MiB Mem : 3921.9 total, 2516.0 free, 894.6 used, 773.1 buff/cache
MiB Swap: 2135.0 total, 2135.0 free, 0.0 used, 3027.3 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR S %CPU  %MEM    TIME+  COMMAND
  689 root        20   0 450556 159200 64668 S 17.3   4.0   1:05.63 Xorg
 2717 hacker      20   0 585232 69940 52868 S  2.7   1.7   0:05.88 qterminal
 1984 hacker      20   0 894372 128388 85996 S  1.0   3.2   0:15.16 xfwm4
 1917 hacker      20   0 215384  3128  2836 S  0.7   0.1   0:17.23 VBoxClient
   641 root        20   0 291740  3652  3264 S  0.3   0.1   0:01.84 VBoxService
 1909 hacker      20   0 214868  3224  2932 S  0.3   0.1   0:03.20 VBoxClient
 1964 hacker      20   0 168744  7504  6864 S  0.3   0.2   0:00.52 at-spi2-registr
 2021 hacker      20   0 275384 26780 19212 S  0.3   0.7   0:00.89 xfsettingsd
 2049 hacker      20   0 222188 47796 19540 S  0.3   1.2   0:12.32 wrapper-2.0
 2135 hacker      20   0 12424  1704  1432 S  0.3   0.0   0:00.16 xcape
    1 root        20   0 23976  14436 10540 S  0.0   0.4   0:01.83 systemd
    2 root        20   0 0 0 0 S  0.0  0.0   0:00.03 kthreadd
    3 root        20   0 0 0 0 S  0.0  0.0   0:00.00 pool_workqueue_release
    4 root        0 -20 0 0 0 I  0.0  0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
    5 root        0 -20 0 0 0 I  0.0  0.0   0:00.00 kworker/R-rcu_gp
    6 root        0 -20 0 0 0 I  0.0  0.0   0:00.00 kworker/R-sync_wq
    7 root        0 -20 0 0 0 I  0.0  0.0   0:00.00 kworker/R-slub_flushwq
    8 root        0 -20 0 0 0 I  0.0  0.0   0:00.00 kworker/R-netns
   11 root        0 -20 0 0 0 I  0.0  0.0   0:00.00 kworker/0:0H-kblockd
   12 root       20   0 0 0 0 I  0.0  0.0   0:00.02 kworker/u8:0-ipv6_addrconf
   13 root       0 -20 0 0 0 I  0.0  0.0   0:00.00 kworker/R-mm_percpu_wq
   14 root       20   0 0 0 0 I  0.0  0.0   0:00.00 rcu_tasks_kthread
   15 root       20   0 0 0 0 I  0.0  0.0   0:00.00 rcu_tasks_rude_kthread
   16 root       20   0 0 0 0 I  0.0  0.0   0:00.00 rcu_tasks_trace_kthread
```

```
hacker@hacker: ~/task1/mytask
(hacker@hacker)~[-/task1/mytask]
$ ps
  PID TTY          TIME CMD
 2761 pts/0    00:00:05 zsh
 23347 pts/0    00:00:00 ps

(hacker@hacker)~[-/task1/mytask]
$ ps aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1   0.0  0.3 23976 14436 ?        Ss   07:30   0:01 /sbin/init splash
root           2   0.0  0.0      0     0 ?        S    07:30   0:00 [kthreadd]
root           3   0.0  0.0      0     0 ?        S    07:30   0:00 [pool_workqueue_release]
root           4   0.0  0.0      0     0 ?        I<   07:30   0:00 [kworker/R-kvfree_rcu_reclaim]
root           5   0.0  0.0      0     0 ?        I<   07:30   0:00 [kworker/R-rcu_gp]
root           6   0.0  0.0      0     0 ?        I<   07:30   0:00 [kworker/R-sync_wq]
root           7   0.0  0.0      0     0 ?        I<   07:30   0:00 [kworker/R-slub_flushwq]
root           8   0.0  0.0      0     0 ?        I<   07:30   0:00 [kworker/R-netns]
root          11   0.0  0.0      0     0 ?        I<   07:30   0:00 [kworker/0:0H-kblockd]
root          12   0.0  0.0      0     0 ?        I    07:30   0:00 [kworker/u8:0-netns]
root          13   0.0  0.0      0     0 ?        I<   07:30   0:00 [kworker/R-mm_percpu_wq]
root          14   0.0  0.0      0     0 ?        I    07:30   0:00 [rcu_tasks_kthread]
root          15   0.0  0.0      0     0 ?        I    07:30   0:00 [rcu_tasks_rude_kthread]
root          16   0.0  0.0      0     0 ?        I    07:30   0:00 [rcu_tasks_trace_kthread]
root          17   0.0  0.0      0     0 ?        S    07:30   0:00 [ksoftirqd/0]
root          18   0.0  0.0      0     0 ?        I    07:30   0:01 [rcu_preempt]
root          19   0.0  0.0      0     0 ?        S    07:30   0:00 [rcu_exp_par_gp_kthread_worker/0]
root          20   0.0  0.0      0     0 ?        S    07:30   0:00 [rcu_exp_gp_kthread_worker]
root          21   0.0  0.0      0     0 ?        S    07:30   0:00 [migration/0]
root          22   0.0  0.0      0     0 ?        S    07:30   0:00 [idle_inject/0]
root          23   0.0  0.0      0     0 ?        S    07:30   0:00 [cpuhp/0]
root          24   0.0  0.0      0     0 ?        S    07:30   0:00 [cpuhp/1]
root          25   0.0  0.0      0     0 ?        S    07:30   0:00 [idle_inject/1]

(hacker@hacker)~[-/task1/mytask]
$ top
top - 08:12
3.4% Tasks: 92, 254 thr, 82 kthr; 1 running
2.9% Load average: 0.15 0.20 0.18
657M/3.83G Uptime: 00:42:22
0K/2.08G

Main I/O
PID USER      PRI NI  VIRT   RES  SHR S  CPU% MEM%  TIME+ Command
689 root        20  0   439M 155M 64668 S  2.8  4.0  1:04.38 /usr/lib/xorg/Xorg :0 -seat seat0 -auth /var/run/lightdm/root/:0 -nolisten tcp vt7 -novtsw
2103 hacker     20  0   358M 44748 31384 S  1.4  1.1  0:01.28 /usr/lib/X86_64-linux-gnu/xfce4/notifyd/xfce4-notifyd
528 root        20  0 18864 9176 7896 S  0.7  0.2  0:00.62 /usr/lib/systemd/systemd-logind
1937 hacker     20  0  210M 3128   0 S  0.7  0.1  0:17.24 /usr/bin/VBoxClient --draganddrop
1984 hacker     20  0  873M 125M 85996 S  0.7  3.2  0:15.23 xfwm4
23664 hacker    20  0  8484 4804 3524 R  0.7  0.1  0:00.25 htop
1 root        20  0 23976 14436 10540 S  0.0  0.4  0:01.84 /sbin/init splash
366 root        20  0 51016 16688 15408 S  0.0  0.4  0:00.55 /usr/lib/systemd/systemd-journald
411 root        20  0 36024 10560 7872 S  0.0  0.3  0:00.22 /usr/lib/systemd/systemd-udev
513 root        20  0  8380 5120 1688 S  0.0  0.1  0:00.53 /usr/sbin/haveged --Foreground --verbose=1
522 root        20  0  301M 7248 6480 S  0.0  0.2  0:00.12 /usr/libexec/accounts-daemon
523 messagebus  20  0 10232 6632 4328 S  0.0  0.2  0:03.63 /usr/bin/dbus-daemon -system --address=systemd: --nofork --nopidfile --systemd-activation
525 polkitd     20  0  373M 9960 7516 S  0.0  0.2  0:00.28 /usr/lib/polkit-1/polkitd --no-debug --log-level=notice
548 root        20  0  301M 7248   0 S  0.0  0.2  0:00.00 /usr/libexec/accounts-daemon
549 root        20  0  301M 7248   0 S  0.0  0.2  0:00.01 /usr/libexec/accounts-daemon
565 root        20  0  301M 7248   0 S  0.0  0.2  0:00.05 /usr/libexec/accounts-daemon
566 root        20  0  328M 19388 16444 S  0.0  0.5  0:00.35 /usr/sbin/NetworkManager --no-daemon
574 polkitd     20  0  373M 9960   0 S  0.0  0.2  0:00.12 /usr/lib/polkit-1/polkitd --no-debug --log-level=notice
576 polkitd     20  0  373M 9960   0 S  0.0  0.2  0:00.00 /usr/lib/polkit-1/polkitd --no-debug --log-level=notice
577 polkitd     20  0  373M 9960   0 S  0.0  0.2  0:00.09 /usr/lib/polkit-1/polkitd --no-debug --log-level=notice
589 root        20  0  381M 12300 10508 S  0.0  0.3  0:00.13 /usr/sbin/ModemManager
597 root        20  0  6864 2840 2584 S  0.0  0.1  0:00.04 /usr/sbin/cron -f
615 root        20  0  381M 12300   0 S  0.0  0.3  0:00.00 /usr/sbin/ModemManager
617 root        20  0  381M 12300   0 S  0.0  0.3  0:00.00 /usr/sbin/ModemManager

F1Help F2Setup F3Search F4Filter F5Tree F6SortBy F7Nice F8Nice F9Kill F10Quit
```

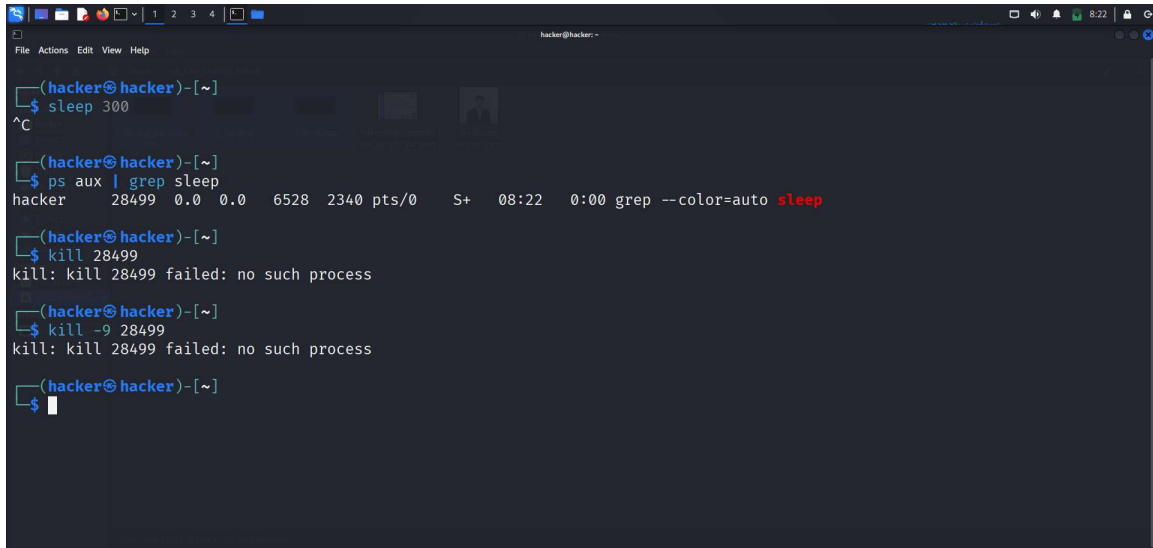
Part B: Process Management

PID found: (Identified using "ps aux | grep sleep")

Command used:

- kill 28499
- kill -9 28499

Result: No such Process.

A terminal window titled 'hacker@hacker: ~' with a menu bar (File, Actions, Edit, View, Help) and a toolbar. The terminal shows the following sequence of commands and output:

```
(hacker@hacker)-[~]  
$ sleep 300  
^C  
(hacker@hacker)-[~]  
$ ps aux | grep sleep  
hacker 28499 0.0 0.0 6528 2340 pts/0 S+ 08:22 0:00 grep --color=auto sleep  
(hacker@hacker)-[~]  
$ kill 28499  
kill: kill 28499 failed: no such process  
(hacker@hacker)-[~]  
$ kill -9 28499  
kill: kill 28499 failed: no such process  
(hacker@hacker)-[~]  
$
```

Part C: System Monitoring

Total RAM: 3.8 GiB

Available RAM: approximately 2.5 GiB

Disk Usage: /dev/sda1 at 79%

System Uptime: about 9 hours

Kernel Version: Kali Linux 6.18.9

Part D: Service Monitoring

Services Checked:

- SSH service: active
- NetworkManager service: Active and running

What is a Service?

- Services are background programs essential for system functionality.

Why are services important?

- They provide essential functionalities (like networking, remote access, logging) continuously without needing direct user interaction.

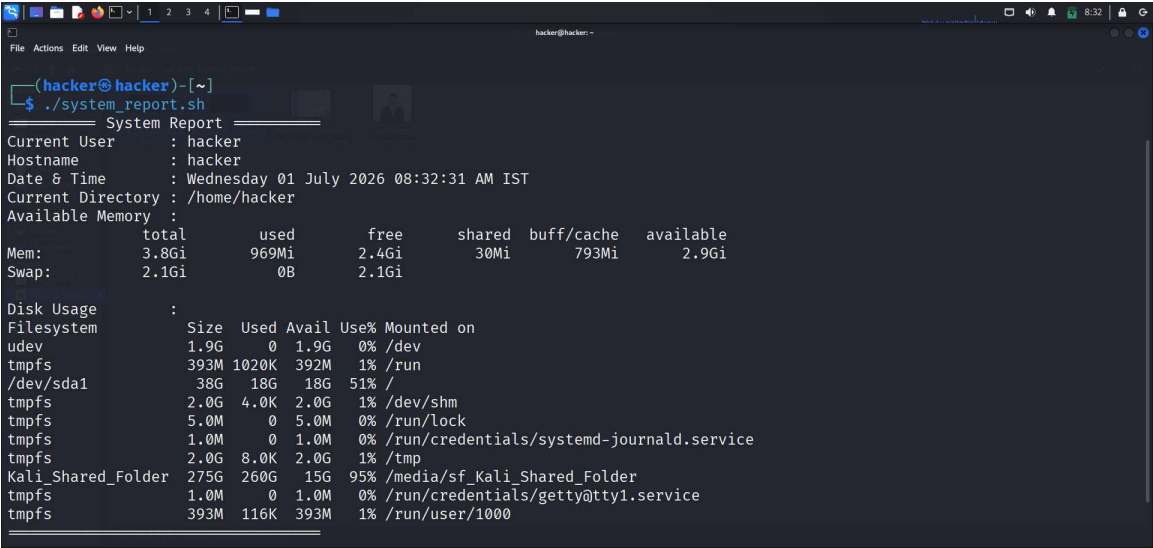
How can a stopped service affect a system?

- If a critical service stops, the functionalities depending on it will fail (e.g., losing internet access if NetworkManager stops).

Part E: Shell Scripting

Script system_report.sh created and executed successfully.

The script displays user, hostname, date, memory usage, and disk usage.



```
(hacker@hacker)~[~]
$ ./system_report.sh
===== System Report =====
Current User      : hacker
Hostname         : hacker
Date & Time      : Wednesday 01 July 2026 08:32:31 AM IST
Current Directory : /home/hacker
Available Memory :
total            used            free            shared    buff/cache    available
Mem:             3.8Gi          969Mi          2.4Gi           30Mi          793Mi          2.9Gi
Swap:            2.1Gi             0B            2.1Gi
Disk Usage      :
Filesystem      Size  Used Avail Use% Mounted on
udev             1.9G   0    1.9G   0% /dev
tmpfs            393M 1020K 392M   1% /run
/dev/sda1        38G   18G   18G  51% /
tmpfs            2.0G  4.0K  2.0G   1% /dev/shm
tmpfs            5.0M   0    5.0M   0% /run/lock
tmpfs            1.0M   0    1.0M   0% /run/credentials/systemd-journald.service
tmpfs            2.0G  8.0K  2.0G   1% /tmp
Kali_Shared_Folder 275G 260G   15G  95% /media/sf_Kali_Shared_Folder
tmpfs            1.0M   0    1.0M   0% /run/credentials/getty@tty1.service
tmpfs            393M 116K  393M   1% /run/user/1000
```

Part F: Security Monitoring Research

Command	Purpose	Example Output	Security Use Case
netstat	network connections	Active Internet connections...	Identifying unauthorized network connections or open listening ports.
ss	socket statistics	State Recv-Q Send-Q Local Address:Port...	Faster alternative to netstat for finding suspicious sockets.
who	logged-in users	hacker tty1 2025-06-15 10:00	Spotting unauthorized users currently logged into the system.
w	user activity	10:00:00 up 9:00, 2 users, load average...	Monitoring current user activity and identifying suspicious commands being run.
last	login history	hacker pts/0 192.168.1.5 Mon Jun 15 09:00 still logged in	Auditing past logins to find anomalous access times or unauthorized IPs.

Part G: Mini SOC Activity

How would you identify resource-heavy processes?

- I would use commands like `top` or `htop` to view real-time system metrics and sort the processes by CPU or Memory usage.

How would you determine whether a process is suspicious?

- I would verify the process by checking its command path, the user running it (owner), its network activity (using `netstat/ss`), and system logs to see if it exhibits anomalous behavior.

What information would you collect before terminating a process?

- Before termination, I would collect the PID, process owner, exact command being executed, resource usage, and any active network connections to preserve evidence for further analysis.